

Supplementary Summer Examination – 2025

Course: B.Tech.

Branch: Civil Engineering

Semester: VI

Subject Code & Name: BTCVE705D_Y20 & Limit State Design of Steel Structures

Max Marks: 60

Date: 25-07-2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.
7. Use of IS 800-2007, IS 800-1984, IS 875 is allowed.

	CO	Marks
Q. 1 Objective type questions. (Compulsory Question)		12
1 The limit state method of design aims to ensure:	CO1	1
a. Minimum weight of structure b. Aesthetic appearance c. Safety and serviceability d. Economy only		
2 The minimum edge distance for a 20 mm bolt as per IS 800 is approximately:	CO1	1
a. 25 mm b. 30 mm c. 34 mm d. 40 mm		
3 The effective length of a column with both ends hinged is:	CO3	1
a. 0.5L b. 1.0L c. 1.2L d. 2.0L		
4 Which of the following does not influence slenderness ratio?	CO3	1
a. Length of member b. Radius of gyration c. Load d. End condition		
5 Which connection is best suited for dynamic or vibrating loads?	CO1	1
a. Riveted b. Friction grip bolts c. Welded d. Ordinary bolts		
6 In which type of failure does the entire cross-section yield uniformly?	CO1	1
a. Shear failure b. Plastic failure c. Brittle failure d. Tension rupture		
7 A tension splice is used when:	CO2	1
a. Two tension members are of different thickness b. Tension member is longer than available length c. To resist moment d. In compression-only members		

8	Which section is most commonly used for gantry girders?	CO2	1
	a. ISMB b. ISMC c. ISWB d. ISHB		
9	The moment of inertia of a built-up section increases when:	CO2	1
	a. Material is removed b. Flange plates are added c. It is laced d. Web is thinned		
10	Gusseted base is used for	CO3	1
	a Light loads. b. Only in timber c. Heavily loaded columns d. Reinforced beams		
11	Block shear failure is observed in:	CO2	1
	a. Bolted connections b. Welded joints c. Column bases d. Gantry girders		
12	Purlins are designed to support:	CO4	1
	a. Columns b. Roof sheeting c. Beams d. Footings		
Q. 2 Solve the following.			12
A)	Explain various types of connections in steel structures and compare bolted and welded joints.	CO1	6
B)	Write short notes on the different types of loads considered in structural steel design.	CO1	6
Q.3 Solve the following.			12
A)	A tension member consists of 2 ISA 100×75×8 mm back to back connected on the same side of a 10 mm thick gusset plate. Design the member to carry a factored load of 250 kN.	CO2	6
B)	Explain the design procedure of a built-up beam using flange plates with a neat sketch.	CO2	6
Q. 4 Solve Any Two of the following.			12
A)	Design a bolted connection to transmit a factored load of 100 kN using 20 mm bolts. Use standard edge and pitch distances.	CO2	6
B)	A 3.0 m long column is made of ISHB 250 @ 537 N/m. Determine its axial load carrying capacity using LSM if it is pinned at both ends. Take $f_y = 250$ MPa.	CO3	6
C)	What are the types of failure in bolted connections? Describe each with diagrams.	CO1	6

Q.5 Solve Any Two of the following.		12
A) Design a purlin to support roofing sheets for an industrial truss spaced 4 m apart. Take $DL = 1.2 \text{ kN/m}^2$, $LL = 1.5 \text{ kN/m}^2$.	CO3	6
B) Describe the design considerations and detailing of a gantry girder.	CO3	6
C) Calculate the dead load per panel point of Howe truss by using following data: Size of industrial shed $16\text{m} \times 32\text{m}$. Spacing of truss is 4m. Weight of GI sheet = 85N/m^2 Weight of purlin = 70 N/m^2 Rise of truss = 2m.	CO3	6
Q. 6 Solve Any Two of the following.		12
A) Design a slab base for a column ISHB 300 carrying a factored axial load of 800 kN. The base rests on M20 concrete.	CO4	6
B) Describe the procedure for designing a grillage foundation with neat sketch.	CO4	6
C) Write short notes on:	CO4	6
i. Column splice		
ii. Anchor bolt design		

*** End ***